



Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Feature Engineering on Financial Ratios

WEEK 3 · DAY 2

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

A feature is something you know, written as a column

A shape

The generator caps maturity at 10 years

An identity

Shared denominators give new ratios

A gap

A missing value carries information

The rule for the whole day

Anything a feature learns from data, it learns from the training rows of each fold, inside a pipeline.

Then two changes that add no column: the quantile transform and clipping.

Course 4's test rows stay closed on both files: every number today is cross-validated inside its training rows.

A feature that writes down a shape

```
oas = BASE_OAS[score] * (0.70 + 0.30 * min(years_to_maturity, 10) / 10) * ...
```

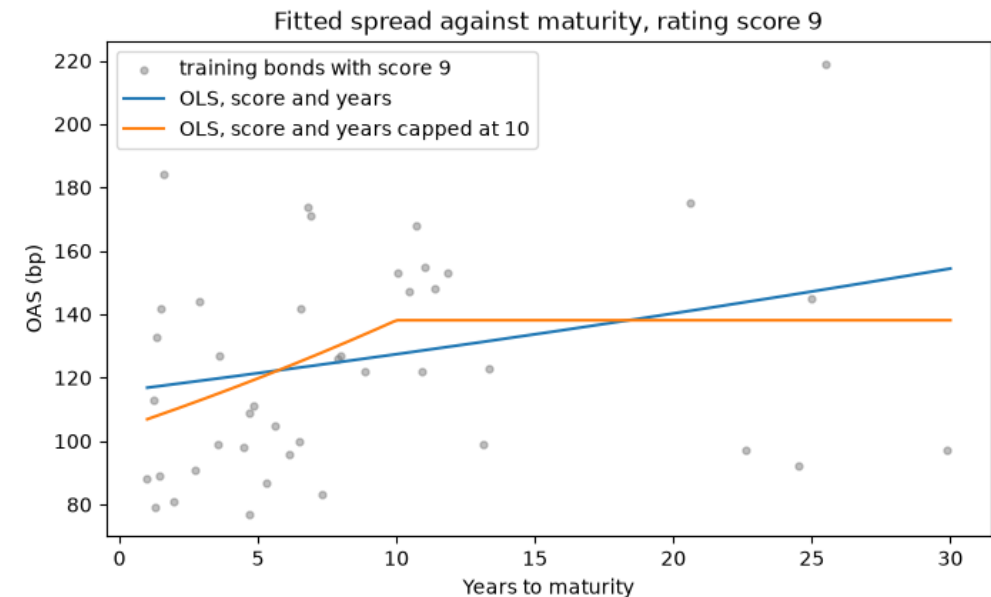
42.3 bp

OLS, score and years

40.9 bp

OLS, score and capped years

181 of the 615 training bonds have more than 10 years left. Cross-validated MAE, five folds of the training bonds.



Same model, same rows, one column written differently. The output is a fitted spread.

With the cap, OLS matches boosting

40.9 bp

OLS, capped years

41.2 bp

gradient boosting, raw years

42.3 bp

OLS, raw years

Two coefficients, a hundred trees

Boosting finds the bend with splits on years. The capped model is handed it, and can be read: one coefficient per column.

What these rows can show

The gap between the first two is far smaller than the spread between one model's best and worst fold.

A feature written from knowledge of the data closes the gap between a simple model and a flexible one.

Ratios from ratios

New feature	Built as	Reads as
debt_to_equity	X2 / X10	total liabilities / equity
ebit_to_liabilities	X7 / X2	EBIT / total liabilities
short_term_share	X51 / X2	short-term share of liabilities

X2 = 0 on 12 training rows

Plain division: 9 infinities, 5 missing. safe_ratio: 0 infinities, 14 missing.

```
ensemblekit.safe_ratio(X_train["Attr7"], X_train["Attr2"])
```

A missing value, a model knows how to handle. An infinity, it does not.

4,680 training rows, 326 bankrupt (6.97 percent). Course 4 reads Attr k as the page's X k.

Ratios as features, 1968

Edward I. Altman, September 1968

"Financial Ratios, Discriminant Analysis and the Prediction of Corporate Bankruptcy" (not read for this course)

The Journal of Finance, volume 23, issue 4, pages 589 to 609

Its title puts financial ratios at the start of a model of company bankruptcy, where today puts them.

Source, read 5 October 2026: the Crossref record of the paper (DOI 10.1111/j.1540-6261.1968.tb00843.x). Bibliographic facts only; the paper itself was not read, and no claim is made about which ratios it used.

Missingness as a feature

Missing ratios	Rows	Bankrupt
0	2,432	3.58%
1	1,698	7.54%
2	272	26.10%
3 or more	278	14.39%

0.7848

LightGBM, raw columns

0.7571

after a median imputer

0.7798

raw, plus the count

Average precision, mean of five folds; fold standard deviation near 0.0448. LightGBM learns which side of each split the gaps go to; an imputer makes a gap look typical.

A description of the training rows only.

The count uses only the row's own cells, so it is a fair feature. The table uses the outcome, so it is not.

Transforms for a linear model

Logistic regression	No indicators	With indicators
Standardized	0.3169	0.5544
Quantile transform	0.5027	0.6926

Average precision, mean of five folds of the training rows

Why it matters here

A straight-line model gives a far value a long lever, and cannot see a gap unless an indicator column says so.

Preparation more than doubles the score; LightGBM on raw columns is still at 0.7848.

Clipping at the training rows' percentiles

X27 runs from -158,130.0 to 214,880.0. Its 1st and 99th percentiles, -77.6614 and 4,516.7060, move 44 values up and as many down; missing values stay missing.

0.5544

standardized, indicators

0.6530

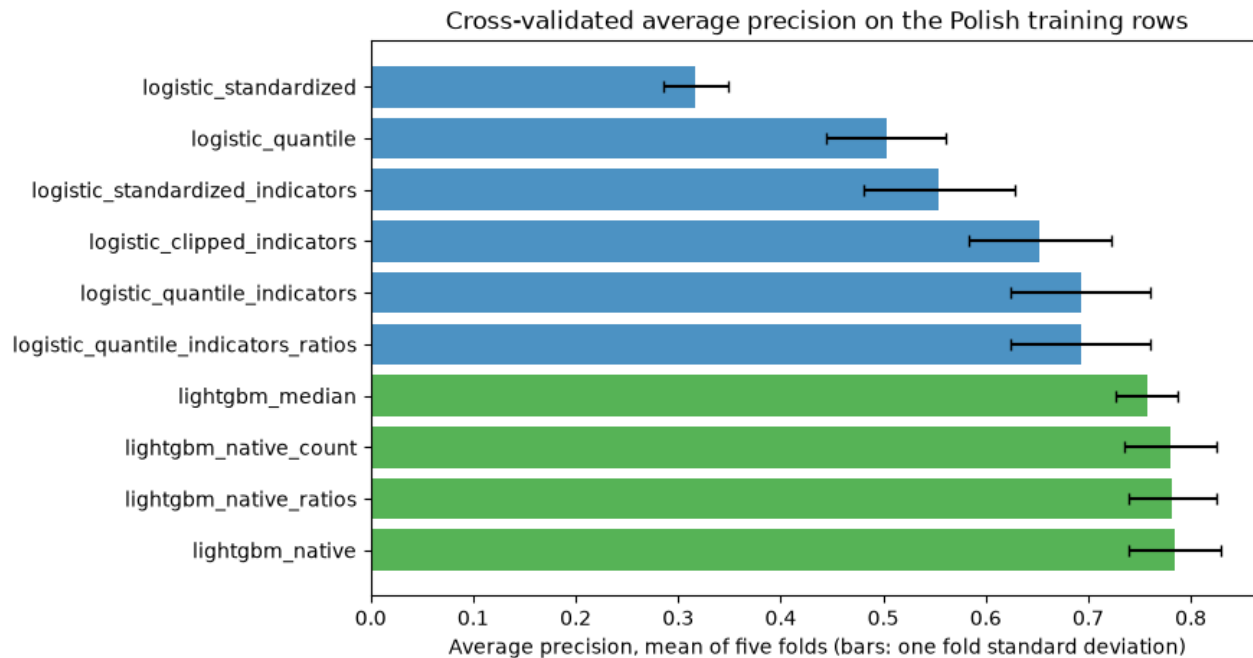
clipped first

X27 bounds	Lower	Upper
fold 1	-71.0499	5345.642
fold 2	-73.4695	4243.157
fold 3	-70.5788	5225.174
fold 4	-138.9016	3832.268
fold 5	-71.0727	4873.028

```
Pipeline([("clip", ensemblekit.ClipToTraining()), ...])
```

Five folds, five sets of bounds: each learned from its own training rows.

What the new ratios add



Model	Without	With ratios
Logistic	0.6926	0.6928
LightGBM	0.7848	0.7821

Both changes are a small fraction of a fold standard deviation: nothing measurable, on this file and these models.

For the linear model the gain came from preparation, not from new columns.

Three kinds of feature, three rules

1. Row by row

The cap, the ratios, the count

Cannot leak. A FunctionTransformer keeps it with the model.

2. Learned from rows

Medians, indicators, quantiles, clip bounds

Inside the pipeline: each fold's training rows only.

3. From the outcome

A bankrupt share by group

The worst leak. Each fold's training rows only, if at all.

And no feature changes day 1's rule: Course 4's test rows are never scored.

Excel side by side

Excel side by side (tested in Excel)

`=MIN(A2,10)`

`=COUNTIF(A2:A616,">10")`

`=AVERAGE(ABS(EXP(C2:C616)-EXP(D2:D616)))`

`=IF(OR(B2=0,B2=""),A2=""),NA(),A2/B2)`

`=A2/B2`

`=PERCENTILE.INC(A2:A4681,0.01), 0.99`

`=MIN(MAX(A2,$D$1),$D$2)`

`=COUNTBLANK(A2:BL2)`

the cap, equal to numpy on 615 bonds

181 bonds changed

40.9 bp, the capped model

#N/A on 14 rows, as safe_ratio

#DIV/0! on 14; 0 for an empty numerator

-77.6614, 4516.7060

an empty cell becomes the lower bound

the missing count, equal on 4,680 rows

Excel has no worksheet function for cross-validation, a pipeline, or boosting. Every feature built today is a formula it has.

Save the day: what the ratios added

1. Code

mlkit.py, ensemblekit.py, its tests

2. Log

Four models; features.txt so far

3. Ratios

Two more models; three names added

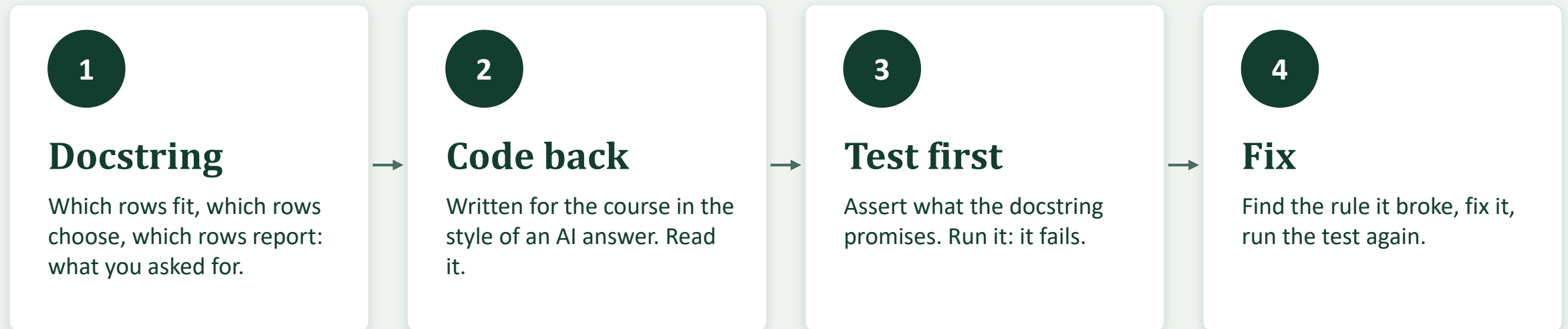
```
git diff HEAD~1 -- features.txt
+debt_to_equity
+ebit_to_liabilities
+short_term_share
```

```
model,rows,roc_auc,average_precision
logistic_quantile_indicators_ratios,cv,0.9208,0.6928
lightgbm_native_ratios,cv,0.9558,0.7821
```

Rows cv: fold means; threshold and flags from the pooled out-of-fold probabilities.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.



Today: a pattern's bankrupt share learned from all rows before the folds: 0.7113, flattered.

Code that runs is not the same as code that is right.

What you can do now

- Write a known shape as a feature
- Build ratios that never return an infinity
- Use gaps: indicators for a linear model
- Transform or clip, learned inside the fold
- Sort every feature into one of three kinds

Week 3, Day 3 Resampling, SMOTE, and class weights